

CHAPTER

5

DIFFERENTIABLE FUNCTIONS

We begin with the definition of the derivative of a function.

Definition 5.1. *Let $I \subset \mathbb{R}$ be an interval and let $x_0 \in I$. We say that $f : I \rightarrow \mathbb{R}$ is differentiable at x_0 or has a derivative at x_0 if*

$$\lim_{x \rightarrow x_0} \frac{f(x) - f(x_0)}{x - x_0}$$

exists. We say that f is differentiable on I if f is differentiable at every point in I .

By definition, f has a derivative at x_0 if there exists a number $L \in \mathbb{R}$ such that for every $\varepsilon > 0$ there exists $\delta > 0$ such that if $|x - x_0| < \delta$ then

$$\left| \frac{f(x) - f(x_0)}{x - x_0} - L \right| < \varepsilon.$$

Derivative function

If f is differentiable at x_0 , we will denote $\lim_{x \rightarrow x_0} \frac{f(x) - f(x_0)}{x - x_0}$ by $f'(x_0)$, that is,

$$f'(x_0) = \lim_{x \rightarrow x_0} \frac{f(x) - f(x_0)}{x - x_0}.$$

The rule that sends x_0 to the number $f'(x_0)$ defines a function on a possibly smaller subset $J \subset I$. The function $f' : J \rightarrow \mathbb{R}$ is called the derivative of f .

Example 5.1. Let $f(x) = 1/x$ for $x \in (0, \infty)$. Prove that $f'(x) = -\frac{1}{x^2}$.

Example 5.2. Let $f(x) = \sin(x)$ for $x \in \mathbb{R}$. Prove that $f'(x) = \cos(x)$.

Solution

Recall that

$$\sin(x) - \sin(x_0) = 2 \sin\left(\frac{x - x_0}{2}\right) \cos\left(\frac{x + x_0}{2}\right)$$

and that $\lim_{x \rightarrow 0} \frac{\sin(x)}{x} = 1$. Therefore,

$$\begin{aligned} \lim_{x \rightarrow x_0} \frac{\sin(x) - \sin(x_0)}{x - x_0} &= \lim_{x \rightarrow x_0} \frac{2 \sin\left(\frac{x - x_0}{2}\right) \cos\left(\frac{x + x_0}{2}\right)}{x - x_0} \\ &= \lim_{x \rightarrow x_0} \left(\frac{\sin\left(\frac{x - x_0}{2}\right)}{\frac{x - x_0}{2}} \right) \cos\left(\frac{x + x_0}{2}\right) \\ &= 1 \cdot \cos(x_0) = \cos(x_0). \end{aligned}$$

Hence $f'(x_0) = \cos(x_0)$ for all x_0 and thus $f'(x) = \cos(x)$.

Example 5.3. Prove by definition that $f(x) = \frac{x}{1+x^2}$ is differentiable on $\mathbb{R}$.

Solution

$$\begin{aligned} \frac{f(x) - f(x_0)}{x - x_0} &= \frac{\frac{x}{1+x^2} - \frac{x_0}{1+x_0^2}}{x - x_0} \\ &= \frac{x(1+x_0^2) - x_0(1+x^2)}{(1+x^2)(1+x_0^2)(x-x_0)} \\ &= \frac{1 - x_0x}{(1+x_0^2)(1+x^2)}. \end{aligned}$$

Now

$$\lim_{x \rightarrow x_0} \frac{f(x) - f(x_0)}{x - x_0} = \frac{1 - x_0^2}{(1 + x_0^2)^2}.$$

Hence, $f'(x_0)$ exists for all $x_0 \in \mathbb{R}$ and the derivative function of f is

$$f'(x) = \frac{1 - x^2}{(1 + x^2)^2}.$$

Example 5.4. Prove that $f'(x) = \alpha$ if $f(x) = \alpha x + b$.

Solution

We have that $f(x) - f(x_0) = \alpha x - \alpha x_0 = \alpha(x - x_0)$. Therefore, $\lim_{x \rightarrow x_0} \frac{f(x) - f(x_0)}{x - x_0} = \alpha$. This proves that $f'(x) = \alpha$ for all $x \in \mathbb{R}$.

Example 5.5. Compute the derivative function of $f(x) = |x|$ for $x \in \mathbb{R}$.

Solution If $x > 0$ then $f(x) = x$ and thus $f'(x) = 1$ for $x > 0$. If $x < 0$ then $f(x) = -x$ and therefore $f'(x) = -1$ for $x < 0$. Now consider $x_0 = 0$. We have that

$$\frac{f(x) - f(x_0)}{x - x_0} = \frac{|x|}{x}.$$

We claim that the limit $\lim_{x \rightarrow 0} \frac{|x|}{x}$ does not exist and thus $f'(0)$ does not exist. To see this, consider $x_n = 1/n$. Then $(x_n) \rightarrow 0$ and $f(x_n) = 1$ for all n . On the other hand, consider $y_n = -1/n$. Then $(y_n) \rightarrow 0$ and $f(y_n) = -1$. Hence, $\lim_{n \rightarrow \infty} f(x_n) \neq \lim_{n \rightarrow \infty} f(y_n)$, and thus the claim holds by the Sequential criterion for limits. The derivative function f' of f is therefore defined on $A = \mathbb{R} \setminus \{0\}$ and is given by

$$f'(x) = \begin{cases} 1, & x > 0 \\ -1, & x < 0. \end{cases}$$

Hence, even though f is continuous at every point in its domain $\mathbb{R}$, it is not differentiable at every point in its domain. In other words, continuity is not a sufficient condition for differentiability.

Geometric interpretation

Let $x_0 \in \mathbb{R}$ and suppose that the function is differentiable at x_0 . Let $T_{(x_0, f(x_0))}$ be the tangent line to the graph of f at the point $(x_0, f(x_0))$, and let (h_n) be a sequence of real numbers decreasing to 0. Let D_n be the straight line joining the points $(x_0, f(x_0))$ and $(x_0 + h_n, f(x_0 + h_n))$. Therefore for all $n \in \mathbb{N}$, $y = \Delta_n(x - x_0) + f(x_0)$, where $\Delta_n = \frac{f(x_0 + h_n) - f(x_0)}{h_n}$, is the equation of D_n . When $n \rightarrow +\infty$, the straight lines D_n tend to the tangent line $T_{(x_0, f(x_0))}$ and the slopes $\Delta_n \rightarrow f'(x_0)$. We conclude that $y = f'(x_0)(x - x_0) + f(x_0)$ is the equation of the tangent line $T_{(x_0, f(x_0))}$. In other words, $f'(x_0)$ is the slope of the tangent line $T_{(x_0, f(x_0))}$.

Right-hand derivative, left-hand derivative

Let $f : I \rightarrow \mathbb{R}$ be function and suppose that there exists $\delta > 0 :]x_0 - \delta, x_0] \subset I$.

Definition 5.2. If $\lim_{x \rightarrow x_0^-} \frac{f(x) - f(x_0)}{x - x_0}$ exists (finite), we say that f is left hand differentiable at x_0 , or has a left hand derivative at x_0 . Noted by $f'(x_0 - 0)$ ($f'_-(x_0)$)

Let $f : I \rightarrow \mathbb{R}$ be function and suppose that there exists $\delta > 0 : [x_0, x_0 + \delta[\subset I$.

Definition 5.3. IF $\lim_{x \rightarrow x_0^+} \frac{f(x) - f(x_0)}{x - x_0}$ exists (finite), we say that f is right hand differentiable at x_0 , or has a right hand derivative at x_0 . Noted by $f'(x_0 + 0)$ ($f'_+(x_0)$).

Remark 5.1. A function f is differentiable at x_0 if and only if both the right-hand derivative and left-hand derivative at x_0 exist and both of these derivatives are equal.

Example 5.6. $f : x \mapsto |x|$ defined on $\mathbb{R}$ is not differentiable at 0.

Theorem 5.1. Suppose that $f : I \rightarrow \mathbb{R}$ is differentiable at x_0 . Then f is continuous at x_0 .

Proof. To prove that f is continuous at x_0 we must show that $\lim_{x \rightarrow x_0} f(x) = f(x_0)$. By assumption $\lim_{x \rightarrow x_0} \frac{f(x) - f(x_0)}{x - x_0} = f'(x_0)$ exists, and clearly $\lim_{x \rightarrow x_0} (x - x_0) = 0$. Hence we can apply the Limits laws and compute

$$\begin{aligned} \lim_{x \rightarrow x_0} f(x) &= \lim_{x \rightarrow x_0} (f(x) - f(x_0) + f(x_0)) \\ &= \lim_{x \rightarrow x_0} \left(\frac{f(x) - f(x_0)}{x - x_0} (x - x_0) + f(x_0) \right) \\ &= f'(x_0) \cdot 0 + f(x_0) \\ &= f(x_0) \end{aligned}$$

and the proof is complete. □

Theorem 5.2. Let $f : I \rightarrow \mathbb{R}$ and $g : I \rightarrow \mathbb{R}$ be differentiable at $x_0 \in I$. The following hold:

- (i) If $\alpha \in \mathbb{R}$ then (αf) is differentiable and $(\alpha f)'(x_0) = \alpha f'(x_0)$.
- (ii) $(f + g)$ is differentiable at x_0 and $(f + g)'(x_0) = f'(x_0) + g'(x_0)$.
- (iii) fg is differentiable at x_0 and $(fg)'(x_0) = f'(x_0)g(x_0) + f(x_0)g'(x_0)$.
- (iv) If $g(x_0) \neq 0$ then (f/g) is differentiable at x_0 and

$$\left(\frac{f}{g} \right)'(x_0) = \frac{f'(x_0)g(x_0) - f(x_0)g'(x_0)}{g(x_0)^2}$$

Proof. Parts (i) and (ii) are straightforward. We will prove only (iii) and (iv). For (iii), we have that

$$\begin{aligned} \frac{f(x)g(x) - f(x_0)g(x_0)}{x - x_0} &= \frac{f(x)g(x) - f(x_0)g(x) + f(x_0)g(x) - f(x_0)g(x_0)}{x - x_0} \\ &= \frac{f(x) - f(x_0)}{x - x_0} g(x) + f(x_0) \frac{g(x) - g(x_0)}{x - x_0}. \end{aligned}$$

Now $\lim_{x \rightarrow x_0} g(x) = g(x_0)$ because g is differentiable at x_0 . Therefore,

$$\begin{aligned} \lim_{x \rightarrow x_0} \frac{f(x)g(x) - f(x_0)g(x_0)}{x - x_0} &= \lim_{x \rightarrow x_0} \frac{f(x) - f(x_0)}{x - x_0} g(x) + \lim_{x \rightarrow x_0} f(x_0) \frac{g(x) - g(x_0)}{x - x_0} \\ &= f'(x_0)g(x_0) + f(x_0)g'(x_0). \end{aligned}$$

To prove part (iv), since $g(x_0) \neq 0$, then there exist a δ -neighborhood $J = (x_0 - \delta, x_0 + \delta)$ such that $g(x) \neq 0$ for all $x \in J$. If $x \in J$ then

$$\begin{aligned} \frac{\frac{f(x)}{g(x)} - \frac{f(x_0)}{g(x_0)}}{x - x_0} &= \frac{\frac{f(x)g(x_0) - g(x)f(x_0)}{g(x)g(x_0)}}{x - x_0} \\ &= \frac{f(x)g(x_0) - f(x_0)g(x_0) + f(x_0)g(x_0) - g(x)f(x_0)}{g(x)g(x_0)(x - x_0)} \\ &= \frac{\frac{f(x)g(x_0) - f(x_0)g(x_0)}{x - x_0} - \frac{f(x_0)g(x) - f(x_0)g(x_0)}{x - x_0}}{g(x)g(x_0)} \end{aligned}$$

Since $g(x_0) \neq 0$, it follows that

$$\lim_{x \rightarrow x_0} \frac{\frac{f(x)}{g(x)} - \frac{f(x_0)}{g(x_0)}}{x - x_0} = \frac{f'(x_0)g(x_0) - f(x_0)g'(x_0)}{g(x_0)^2}$$

and the proof is complete. □

We now move to the Chain Rule.

Theorem 5.3. *Let $f : I \rightarrow \mathbb{R}$ and $g : J \rightarrow \mathbb{R}$ be functions such that $f(I) \subset J$ and let $x_0 \in I$. If $f'(x_0)$ exists and $g'(f(x_0))$ exists then $(g \circ f)'(x_0)$ exists and $(g \circ f)'(x_0) = g'(f(x_0))f'(x_0)$.*

Proof. Suppose that there exists a neighborhood of x_0 where $f(x) \neq f(x_0)$. Otherwise, the composite function $(g \circ f)(x)$ is constant in a neighborhood of x_0 , and then clearly differentiable at x_0 . Consider the function $h : J \rightarrow \mathbb{R}$ defined by

$$h(y) = \begin{cases} \frac{g(y) - g(f(x_0))}{y - f(x_0)}, & y \neq f(x_0) \\ g'(f(x_0)), & y = f(x_0). \end{cases}$$

Now

$$\begin{aligned} \lim_{y \rightarrow f(x_0)} h(y) &= \lim_{y \rightarrow f(x_0)} \frac{g(y) - g(f(x_0))}{y - f(x_0)} \\ &= g'(f(x_0))' \\ &= h(f(x_0)). \end{aligned}$$

Hence, h is differentiable at $f(x_0)$ and therefore h is at $f(x_0)$. Now,

$$\frac{g(f(x)) - g(f(x_0))}{x - x_0} = h(f(x)) \frac{f(x) - f(x_0)}{x - x_0}$$

and therefore

$$\begin{aligned} \lim_{x \rightarrow x_0} \frac{g(f(x)) - g(f(x_0))}{x - x_0} &= \lim_{x \rightarrow x_0} h(f(x)) \frac{f(x) - f(x_0)}{x - x_0} \\ &= h(f(x_0)) f'(x_0) \\ &= g'(f(x_0)) f'(x_0). \end{aligned}$$

Therefore, $(g \circ f)'(x_0) = g'(f(x_0)) f'(x_0)$ as claimed.

□

Example 5.7. Compute $f'(x)$ if

$$f(x) = \begin{cases} x^2 \sin\left(\frac{1}{x}\right), & x \neq 0 \\ 0, & x = 0. \end{cases}$$

Where is $f'(x)$ continuous?

Solution

When $x \neq 0$, $f(x)$ is the composition and product of differentiable functions at x , and therefore f is differentiable at $x \neq 0$. For instance, on $A = \mathbb{R} \setminus \{0\}$, the functions $1/x$, $\sin(x)$ and x^2 are differentiable at every $x \in A$. Hence, if $x \neq 0$ we have that

$$f'(x) = 2x \sin\left(\frac{1}{x}\right) - \cos\left(\frac{1}{x}\right).$$

Consider now $x_0 = 0$. If $f'(0)$ exists it is equal to

$$\begin{aligned} \lim_{x \rightarrow 0} \frac{f(x) - f(x_0)}{x - x_0} &= \lim_{x \rightarrow 0} \frac{x^2 \sin\left(\frac{1}{x}\right)}{x} \\ &= \lim_{x \rightarrow 0} x \sin\left(\frac{1}{x}\right). \end{aligned}$$

Using the Squeeze Theorem, we deduce that $f'(0) = 0$. Therefore,

$$f'(x) = \begin{cases} 2x \sin\left(\frac{1}{x}\right) - \cos\left(\frac{1}{x}\right), & x \neq 0 \\ 0, & x = 0. \end{cases}$$

From the above formula obtained for $f'(x)$, we observe that when $x \neq 0$ f' is continuous since it is the product/difference/composition of continuous functions. To determine

continuity of f' at $x = 0$ consider $\lim_{x \rightarrow 0} f'(x)$. Consider the sequence $x_n = \frac{1}{n\pi}$, which clearly converges to $x_0 = 0$. Now, $f'(x_n) = \frac{2}{n\pi} \sin(n\pi) - \cos(n\pi)$. Now, $\sin(n\pi) = 0$ for all n and therefore $f'(x_n) = -\cos(n\pi) = (-1)^{n+1}$. The sequence $f'(x_n)$ does not converge and therefore $\lim_{x \rightarrow 0} f'(x)$ does not exist. Thus, f' is not continuous at $x = 0$.

Example 5.8. Compute $f'(x)$ if

$$f(x) = \begin{cases} x^3 \sin\left(\frac{1}{x}\right), & x \neq 0 \\ 0, & x = 0 \end{cases}$$

Where is $f'(x)$ continuous?

Solution

When $x \neq 0$, $f(x)$ is the composition and product of differentiable functions, and therefore f is differentiable at $x \neq 0$. For instance, on $A = \mathbb{R} \setminus \{0\}$, the functions $1/x$, $\sin(x)$ and x^3 are differentiable on A . Hence, if $x \neq 0$ we have that

$$f'(x) = 3x^2 \sin\left(\frac{1}{x}\right) - x \cos\left(\frac{1}{x}\right).$$

Consider now $x_0 = 0$. If $f'(0)$ exists it is equal to

$$\begin{aligned} \lim_{x \rightarrow 0} \frac{f(x) - f(x_0)}{x - x_0} &= \lim_{x \rightarrow 0} \frac{x^3 \sin\left(\frac{1}{x}\right)}{x} \\ &= \lim_{x \rightarrow 0} x^2 \sin\left(\frac{1}{x}\right) \end{aligned}$$

and using the Squeeze Theorem we deduce that $f'(0) = 0$. Therefore,

$$f'(x) = \begin{cases} 3x^2 \sin\left(\frac{1}{x}\right) - x \cos\left(\frac{1}{x}\right), & x \neq 0 \\ 0, & x = 0. \end{cases}$$

When $x \neq 0$, f' is continuous since it is the product/difference/composition of continuous functions. To determine continuity of f' at $x_0 = 0$ we consider the limit $\lim_{x \rightarrow 0} f'(x)$. Now $\lim_{x \rightarrow 0} 3x^2 \sin\left(\frac{1}{x}\right) = 0$ using the Squeeze Theorem, and similarly $\lim_{x \rightarrow 0} x \cos\left(\frac{1}{x}\right) = 0$ using the Squeeze Theorem. Therefore, $\lim_{x \rightarrow 0} f'(x)$ exists and is equal to 0, which is equal to $f'(0)$. Hence, f' is continuous at $x = 0$, and thus continuous everywhere.

Example 5.9. Consider the function

$$f(x) = \begin{cases} x^2 \sin\left(\frac{1}{x}\right), & x \in \mathbb{Q} \setminus \{0\} \\ x^2 \cos\left(\frac{1}{x}\right), & x \notin \mathbb{Q} \\ 0, & x = 0. \end{cases}$$

Show that $f'(0) = 0$.

5.0.1 Some theorems

Definition 5.4. Let $f : I \rightarrow \mathbb{R}$ be a function and let $x_0 \in I$.

(i) We say that f has a relative maximum at x_0 if there exists $\delta > 0$ such that $f(x) \leq f(x_0)$ for all $x \in (x_0 - \delta, x_0 + \delta)$.

(ii) We say that f has a relative minimum at x_0 if there exists δ such that $f(x_0) \leq f(x)$ for all $x \in (x_0 - \delta, x_0 + \delta)$.

A point $x_0 \in I$ is called a critical point of $f : I \rightarrow \mathbb{R}$ if $f'(x_0) = 0$. The next theorem says that a relative maximum/minimum of a differentiable function can only occur at a critical point.

Theorem 5.4. (Pierre Fermat: 1601-1665) Let $f : I \rightarrow \mathbb{R}$ be a function and let x_0 be an interior point of I . Suppose that f has a relative maximum (or minimum) at x_0 . If f is differentiable at x_0 then x_0 is a critical point of f , that is, $f'(x_0) = 0$.

Proof. Suppose that f has a relative maximum at x_0 ; the relative minimum case is similar. Then for $x \neq x_0$, it holds that $f(x) - f(x_0) \leq 0$ for $x \in (x_0 - \delta, x_0 + \delta)$ and some $\delta > 0$. Consider the function $h : (x_0 - \delta, x_0 + \delta) \rightarrow \mathbb{R}$ defined by

$$h(x) = \begin{cases} \frac{f(x) - f(x_0)}{x - x_0}, & x \neq x_0 \\ f'(x_0), & x = x_0 \end{cases}$$

Then the function h is continuous at $x_0 = 0$ because $\lim_{x \rightarrow x_0} h(x) = h(x_0)$. Now for $x \in A = (x_0, x_0 + \delta)$ it holds that $h(x) \leq 0$ and therefore $f'(x_0) = \lim_{x \rightarrow x_0} h(x) \leq 0$. Similarly, for $x \in B = (x_0 - \delta, x_0)$ it holds that $h(x) \geq 0$ and therefore $0 \leq f'(x_0)$. Thus $f'(x_0) = 0$. $\square$

Corollary 5.1. If $f : I \rightarrow \mathbb{R}$ has a relative maximum (or minimum) at x_0 then either $f'(x_0) = 0$ or $f'(x_0)$ does not exist.

Example 5.10. The function $f(x) = |x|$ has a relative minimum at $x = 0$, however, f is not differentiable at $x = 0$.

Theorem 5.5. (Michele Rolle: 1652-1719) Let $f : [a, b] \rightarrow \mathbb{R}$ be continuous on $[a, b]$ and differentiable on $]a, b[$. If $f(a) = f(b)$ then there exists $x_0 \in (a, b)$ such that $f'(x_0) = 0$.

Proof. Since f is continuous on $[a, b]$ it achieves its maximum and minimum at some point x^* and x_* , respectively, that is $f(x_*) \leq f(x) \leq f(x^*)$ for all $x \in [a, b]$. If f is constant then $f'(x) = 0$ for all $x \in (a, b)$. If f is not constant then $f(x_*) < f(x^*)$. Since $f(a) = f(b)$ it follows that at least one of x_* and x^* is not contained in $\{a, b\}$, and hence there exists $x_0 \in \{x_*, x^*\}$ such that $f'(x_0) = 0$. $\square$

Remark 5.2. Rolle remains true even when the interval is open, provided that $\lim_{x \rightarrow a^+} f(x) = \lim_{x \rightarrow b^-} f(x)$.

We now state and prove the main result of this section.

Theorem 5.6. (Mean Value theorem: Lagrange 1736-1813) Let $f : [a, b] \rightarrow \mathbb{R}$ be continuous on $[a, b]$ and differentiable on (a, b) . Then there exists $x_0 \in (a, b)$ such that $f'(x_0) = \frac{f(b) - f(a)}{b - a}$.

Proof. If $f(a) = f(b)$ then the result follows from Rolle's Theorem ($f'(x_0) = 0$ for some $x_0 \in (a, b)$). Let $h : [a, b] \rightarrow \mathbb{R}$ be the line from $(a, f(a))$ to $(b, f(b))$, that is,

$$h(x) = f(a) + \frac{f(b) - f(a)}{(b - a)}(x - a)$$

and define the function

$$g(x) = f(x) - h(x)$$

for $x \in [a, b]$. Then $g(a) = f(a) - f(a) = 0$ and $g(b) = f(b) - f(b) = 0$, and thus $g(a) = g(b)$. Clearly, g is continuous on $[a, b]$ and differentiable on (a, b) , and it is straightforward to verify that $g'(x) = f'(x) - \frac{f(b) - f(a)}{b - a}$. By Rolle's Theorem, there exists $x_0 \in (a, b)$ such that $g'(x_0) = 0$, and therefore $f'(x_0) = \frac{f(b) - f(a)}{b - a}$.

Theorem 5.7. Extended mean value theorem: Cauchy theorem 1789-1857)

Let f, g be two continuous functions on $[a, b]$ and differentiable on $]a, b[$. If g' does not vanish on $]a, b[$, then there exists a point $x_0 \in]a, b[$ such that $\frac{f(b) - f(a)}{g(b) - g(a)} = \frac{f'(x_0)}{g'(x_0)}$.

Proof. Consider the function

$$h(x) = (f(x) - f(a))(g(b) - g(a)) - (f(b) - f(a))(g(x) - g(a)).$$

This is continuous on $[a, b]$ and differentiable on (a, b) , with

$$h'(x) = f'(x)(g(b) - g(a)) - g'(x)(f(b) - f(a)).$$

Note that $h(a) = 0 = h(b)$. By Rolle's Theorem, there a spot x_0 where $h'(x_0) = 0$. $\square$

Remark 5.3. *Lagrange theorem is a special case of Cauchy theorem: $g(x) = x, x \in [a, b]$.*

Example 5.11. *Prove that $\forall x > 0 : e^x > 1 + x + \frac{x^2}{2}$.*

Proof. Cauchy theorem applied with $f(x) = e^x, g(x) = 1 + x + \frac{x^2}{2}, x \in [0, x]$. Then

$$\exists x_0 \in]0, x[: \frac{e^x - e^0}{1 + x + \frac{x^2}{2} - 1} = \frac{e^{x_0}}{1 + x_0} \text{ but } \frac{e^{x_0}}{1 + x_0} > 1, \forall x_0 > 0.$$

So

$$\frac{e^x - 1}{x + \frac{x^2}{2}} > 1 \Rightarrow e^x > 1 + x + \frac{x^2}{2}.$$

$\square$

L'Hospital's Rule And Indeterminate Forms

L'Hospital's Rule (Guillaume L'Hospital (1661-1704)) tells us that if we have an indeterminate form $\frac{0}{0}$ or $\frac{\infty}{\infty}$, all we need to do is differentiate the numerator and differentiate the denominator and then take the limit.

Theorem 5.8. *Let $f(x)$ and $g(x)$ be continuous functions on an interval containing $x = a$, with $f(a) = g(a) = 0$. Suppose that f and g are differentiable, and that f' and g' are continuous. Finally, suppose that $g'(a) \neq 0$. Then*

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)} = \frac{f'(a)}{g'(a)}.$$

Also,

$$\lim_{x \rightarrow a^+} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a^+} \frac{f'(x)}{g'(x)}$$

and

$$\lim_{x \rightarrow a^-} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a^-} \frac{f'(x)}{g'(x)}$$

Proof. Since that $f(a) = g(a) = 0$ and $g'(a) \neq 0$. Then, for any x , $f(x) = f(x) - f(a)$ and $g(x) = g(x) - g(a)$. But then,

$$\begin{aligned}\lim_{x \rightarrow a} \frac{f(x)}{g(x)} &= \lim_{x \rightarrow a} \frac{f(x) - f(a)}{g(x) - g(a)} \\ &= \lim_{x \rightarrow a} \frac{[f(x) - f(a)]/(x - a)}{[g(x) - g(a)]/(x - a)} \\ &= \frac{\lim_{x \rightarrow a} ([f(x) - f(a)]/(x - a))}{\lim_{x \rightarrow a} ([g(x) - g(a)]/(x - a))} \\ &= \frac{f'(a)}{g'(a)},\end{aligned}$$

since, by definition, $f'(a) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$ and $g'(a) = \lim_{x \rightarrow a} \frac{g(x) - g(a)}{x - a}$. Since f' and g' are assumed to be continuous, this is also

$$\frac{\lim_{x \rightarrow a} f'(x)}{\lim_{x \rightarrow a} g'(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}.$$

□

This version is easy to prove, and is good enough to compute limits like

$$\lim_{x \rightarrow 0} \frac{\sin(2x)}{x + x^2}.$$

However, it isn't good enough to compute limits like

$$\lim_{x \rightarrow 0} \frac{1 - \cos(2x)}{x^2}$$

since in that case $g'(0) = 0$. To solve problems like the last one, we need the following version.

Theorem 5.9. *Suppose that f and g are continuous on a closed interval $[a, b]$, and are differentiable on the open interval (a, b) . Suppose that $g'(x)$ is never zero on (a, b) , and that $\lim_{x \rightarrow a^+} f'(x)/g'(x)$ exists, and that $\lim_{x \rightarrow a^+} f(x) = \lim_{x \rightarrow a^+} g(x) = 0$. Then*

$$\lim_{x \rightarrow a^+} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a^+} \frac{f'(x)}{g'(x)}.$$

Proof. By assumption, f and g are differentiable to the right of a , and the limits of f and g as $x \rightarrow a^+$ are zero. Define $f(a)$ to be zero, and likewise define $g(a) = 0$. Since these values agree with the limits, f and g are continuous on some half-open interval $[a, b)$ and differentiable on (a, b) .

For any $x \in (a, b)$, we have that f and g are differentiable on (a, x) and continuous on $[a, x]$. By the extended MVT, there is a point c between a and x such that $f'(c)g(x) =$

$f'(x)g(c)$. In other words, $f'(c)/g'(c) = f(x)/g(x)$. Also, as x approaches a , c also approaches a , since c is somewhere between x and a . But then

$$\lim_{x \rightarrow a^+} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a^+} \frac{f'(c)}{g'(c)} = \lim_{c \rightarrow a^+} \frac{f'(c)}{g'(c)}$$

That last expression is the same as $\lim_{x \rightarrow a^+} f'(x)/g'(x)$. □

Note that this theorem doesn't require anything about $g'(a)$, just about how g' behaves to the right of a . An analogous theorem applies to the limit as $x \rightarrow a^-$ (and requires f and g and f' and g' to be defined on an interval that ends at a , rather than one that starts at a). You can combine the two to get a theorem about an overall limit as $x \rightarrow a$.

The conclusion of L'Hôpital's Rule relates one limit (of f/g) to another limit (of f'/g'), and not to the value of $f'(a)/g'(a)$. This is what allows the theorem to be used recursively to solve problems.

The inverse is not always true, see the following

Example 5.12. We have $\lim_{x \rightarrow 0} \frac{x^2 \sin \frac{1}{x}}{\sin x} = \lim_{x \rightarrow 0} \frac{x^2 \sin \frac{1}{x}}{x} = \lim_{x \rightarrow 0} x \sin \frac{1}{x} = 0$,
 despite that $\lim_{x \rightarrow 0} \frac{\left(x^2 \sin \frac{1}{x}\right)'}{(\sin x)'} = \lim_{x \rightarrow 0} \frac{2x \sin \frac{1}{x} - \cos \frac{1}{x}}{\cos x}$ does not exist.

Remark 5.4. If $\frac{f'(x)}{g'(x)}$ is an indeterminate form such $\frac{0}{0}$ and we suppose that $f'(x)$ and $g'(x)$ satisfy L'Hôpital theorem hypothesis, then $\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)} = \lim_{x \rightarrow a} \frac{f''(x)}{g''(x)}$.

Theorem 5.10. Extended L'Hôpital's Rule Suppose that f and g are two functions well defined and differentiable on a neighbourhood of a point a , may be except at a and assume that $\lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a} g(x) = \infty$, and that $g(x) \neq 0$ et $g'(x) \neq 0$ on the candidate neighbourhood. So, if $\lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}$ exists, then $\lim_{x \rightarrow a} \frac{f(x)}{g(x)}$ exists and $\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}$.

Remark 5.5. If $a = \infty$ the transformation $x = \frac{1}{t}$ guide us to the case $a = 0$,

$$\lim_{x \rightarrow \infty} \frac{f(x)}{g(x)} = \lim_{t \rightarrow 0} \frac{f\left(\frac{1}{t}\right)}{g\left(\frac{1}{t}\right)} = \lim_{t \rightarrow 0} \frac{\left(f\left(\frac{1}{t}\right)\right)'}{\left(g\left(\frac{1}{t}\right)\right)'} = \lim_{t \rightarrow 0} \frac{-\frac{1}{t^2} f'\left(\frac{1}{t}\right)}{-\frac{1}{t^2} g'\left(\frac{1}{t}\right)} = \lim_{x \rightarrow \infty} \frac{f'(x)}{g'(x)}.$$

Theorem 5.11. Let $f : [a, b] \rightarrow \mathbb{R}$ be continuous on $[a, b]$ and differentiable on (a, b) . If $f'(x) = 0$ for all $x \in (a, b)$ then f is constant on $[a, b]$.

Proof. Let $y \in (a, b]$. Now f restricted to $[a, y]$ satisfies all the assumptions needed in the Mean Value Theorem. Therefore, there exists $x_0 \in (a, y)$ such that $f'(x_0) = \frac{f(y)-f(a)}{y-a}$. But $f'(x_0) = 0$ and thus $f(y) = f(a)$. This holds for all $y \in (a, b]$ and thus f is constant on $[a, b]$ $\square$

Corollary 5.2. If $f, g : [a, b] \rightarrow \mathbb{R}$ are continuous and differentiable on (a, b) and $f'(x) = g'(x)$ for all $x \in (a, b)$ then $f(x) = g(x) + C$ for some constant C .

The sign of the derivative f' determines where f is increasing or decreasing.

Theorem 5.12. Suppose that $f : I \rightarrow \mathbb{R}$ is differentiable.

- (i) Then f is increasing if and only if $f'(x) \geq 0$ for all $x \in I$.
- (ii) Then f is decreasing if and only if $f'(x) \leq 0$ for all $x \in I$.

Proof. Suppose that f is increasing. Then for all $x, x_0 \in I$ with $x \neq x_0$ it holds that $\frac{f(x)-f(x_0)}{x-x_0} \geq 0$ and therefore $f'(x_0) = \lim_{x \rightarrow x_0} \frac{f(x)-f(x_0)}{x-x_0} \geq 0$. Hence, this proves that $f'(x) \geq 0$ for all $x \in I$.

Now suppose that $f'(x) \geq 0$ for all $x \in I$. Suppose that $x < y$. Then by the Mean Value Theorem, there exists $x_0 \in (x, y)$ such that $f'(x_0) = \frac{f(y)-f(x)}{y-x}$. Therefore, since $f'(x_0) \geq 0$ it follows that $f(y) - f(x) \geq 0$.

Part (ii) is proved similarly. $\square$

Derivative of inverse functions

Theorem 5.13. Let $f :]a, b[\rightarrow \mathbb{R}$, $-\infty \leq a < b \leq +\infty$ be a function that is both invertible and differentiable. Let $f(]a, b[) =]c, d[$ ($-\infty \leq c < d \leq +\infty$) et $f^{-1} :]c, d[\rightarrow]a, b[$, the inverse function of f . If f is differentiable at x_0 such that $f'(x_0) \neq 0$, then f^{-1} is differentiable at $f(x_0)$ and satisfies $(f^{-1})'(f(x_0)) = \frac{1}{f'(x_0)}$.

Proof. Use composition derivative formula. $\square$

5.1 Higher order derivatives

Let $f :]a, b[\rightarrow \mathbb{R}$ be a differentiable function on $]a, b[$. Let $g = f' :]a, b[\rightarrow \mathbb{R}$ be the derivative function by setting $g(x) = f'(x), \forall x \in]a, b[$.

Definition 5.5. At $x_0 \in]a, b[$ the derivative of the derivative of a function f if it exists, is called the second derivative of that function and is denoted by one of the symbols $f''(x_0)$, $\frac{d^2}{dx^2}f(x_0)$.

If the derivative of $n \in \mathbb{N}$ exists, then, we denote it by $f^{(n)}(x_0)$ or $\frac{d^n}{dx^n}f(x_0)$ and if for all $x \in]a, b[$, $f^{(n)}(x)$ exists, then the derivative of order $n + 1$ is defined by $f^{(n+1)}(x) = (f^{(n)}(x))'$, if it exists.

Example 5.13. $\forall n \in \mathbb{N}, \forall x \in \mathbb{R}$,

$$(\sin x)^{(n)} = \sin\left(x + n\frac{\pi}{2}\right), (\cos x)^{(n)} = \cos\left(x + n\frac{\pi}{2}\right)$$

By induction we have

For $n = 1$, one has $(\sin x)' = \cos x = \sin\left(x + \frac{\pi}{2}\right)$, is true.

Suppose that it is true for n .

So $(\sin x)^{(n+1)} = (\sin^{(n)} x)' \left(\sin\left(x + n\frac{\pi}{2}\right)\right)' = \cos\left(x + n\frac{\pi}{2}\right) = \sin\left(x + n\frac{\pi}{2} + \frac{\pi}{2}\right) = \sin\left(x + (n+1)\frac{\pi}{2}\right)$
which is true too.

Technical fact

If $y = f(x)$ is a function of time that describes the position of a moving object, then:

1. The first derivative represents the velocity of the object.
2. The second derivative represents the acceleration of the object.
3. The third derivative represents the jerk of the object.

Definition 5.6. Let $A \subset \mathbb{R}$. Then $f \in C^{(n)}(A)$, $(n \in \mathbb{N}) \iff \forall x \in A, f^{(n)}(x) (n \in \mathbb{N})$ exists and $f^{(n)} \in C(A) (n \in \mathbb{N})$. f is said to be n times continuously differentiable.

Definition 5.7. $C^{(\infty)}(A) = \bigcap_{n=1}^{\infty} C^{(n)}(A)$

$f \in C^{(\infty)}(A) \iff \forall n \in \mathbb{N} : f \in C^{(n)}(A)$, f is said to be infinitely differentiable.

Example 5.14. Functions $\sin x, \cos x, e^x, x \in \mathbb{R}$ belong to $C^{(\infty)}(\mathbb{R})$.

Theorem 5.14. (Leibniz formula) Let $\{f, g\} \subset C^{(n)}(]a, b[)$ for $n \in \mathbb{N}$. Then $fg \in C^{(n)}(]a, b[)$, and we have

$$(fg)^{(n)} = \sum_{k=0}^n C_n^k f^{(k)} g^{(n-k)}.$$

Proof. By induction. □

Example 5.15. Calculate $(x^2 \sin 2x)^{(10)}$. One has $f(x) = x^2$, $g(x) = \sin 2x$ and $f'(x) = 2x$, $f''(x) = 2$, $f'''(x) = \dots = f^{(10)}(x) = 0$

$$\begin{aligned} g'(x) &= 2 \sin\left(2x + \frac{\pi}{2}\right) = 2 \cos 2x, \quad g''(x) = 2^2 \sin\left(2x + 2\frac{\pi}{2}\right) = 2^2 (-\sin 2x), \dots, \\ g^{(10)}(x) &= 2^{10} \sin\left(2x + 10\frac{\pi}{2}\right) = 2^{10} \sin(2x + 5\pi) = -2^{10} \sin 2x \end{aligned}$$

So

$$\begin{aligned} (x^2 \sin 2x)^{(10)} &= f^{(0)}g^{(10)} + C_{10}^1 f'g^{(9)} + C_{10}^2 f''g^{(8)} + C_{10}^3 f'''g^{(7)} \\ &= -2^{10}x^2 \sin 2x + 10(2x)2^9 \sin\left(2x + 9\frac{\pi}{2}\right) + \frac{10 \cdot 9}{2} \sin\left(2x + 8\frac{\pi}{2}\right) \\ &= -2^{10}x^2 \sin 2x + 2^{10}10x \cos 2x + 8^8 \cdot 10 \cdot 9 \sin 2x \\ &= -2^{10} \left(x^2 \sin 2x - 10x \cos 2x - \frac{45}{2} \sin 2x \right). \end{aligned}$$

Local and absolute extrema

An extremum (or extreme value) of a function is a point at which a maximum or minimum value of the function is obtained in some interval. A local extremum (or relative extremum) of a function is the point at which a maximum or minimum value of the function in some open interval containing the point is obtained.

An absolute extremum (or global extremum) of a function in a given interval is the point at which a maximum or minimum value of the function is obtained. Frequently, the interval given is the function's domain, and the absolute extremum is the point corresponding to the maximum or minimum value of the entire function.

Definition 5.8. Let $x_0 \in I$ and let $f : I \rightarrow \mathbb{R}$ be a function, where I is a real interval.

We say that f reaches at x_0 a local maximum if there exists $\eta > 0$ such that $[x_0 - \eta, x_0 + \eta] \subset I$ and for all $x \in [x_0 - \eta, x_0 + \eta]$ we have $f(x) \leq f(x_0)$.

We say that f reaches at x_0 a local minimum if there exists $\eta > 0$ such that $[x_0 - \eta, x_0 + \eta] \subset I$ and for all $x \in [x_0 - \eta, x_0 + \eta]$ we have $f(x) \geq f(x_0)$. We say that f reaches at x_0 a local extremal value if f reaches at x_0 a local maximum or minimum.

Theorem 5.15. If $f : I \rightarrow \mathbb{R}$ reaches at $x_0 \in I$ a local extremal value and f is differentiable at x_0 , then $f'(x_0) = 0$.

Proof. If f reaches at x_0 a local maximum, then

$$f'(x_0) = f'(x_0^+) = \lim_{x \rightarrow x_0} \frac{f(x) - f(x_0)}{x - x_0} \leq 0$$

and

$$f'(x_0) = f'(x_0^-) = \lim_{x \rightarrow x_0} \frac{f(x) - f(x_0)}{x - x_0} \geq 0$$

Hence $f'(x_0) = 0$. □

Remark 5.6. The reciprocal is false since the function $f(x) = x^3$ has $f'(0) = 0$ but it does not reach an extremal value at 0.

Theorem 5.16. Suppose that x_0 is a stationary (critical) point of a given function f ($i, e : f'(x) = 0$) and suppose that the second derivative of f is continuous in some neighbourhood of x_0 .

If $f''(x_0) < 0$, then f admits a local maximum local at x_0 ;

If $f''(x_0) > 0$, then f admits a local minimum local at x_0 .

Theorem 5.17. Suppose that $f'(x_0) = f''(x_0) = \dots = f^{(n)}(x_0) = 0$ and suppose that $f^{(n+1)}(x) \neq 0$ is continuous in some neighbourhood of x_0 . If $n + 1$ is even and $f^{(n+1)}(x_0) < 0$, then f admits a local maximum at x_0 ; If $n + 1$ is even and $f^{(n+1)}(x_0) > 0$, then f admits a local minimum at x_0 ; If $n + 1$ is odd, then f does not have any local extrema at x_0 .

Example 5.16. Let be the function $f(x) = e^x + e^{-x} + 2 \cos x$, $f'(x) = e^x - e^{-x} - 2 \sin x$, remark that $x = 0$ is a stationary point. Then

$$f''(x) = e^x + e^{-x} - 2 \cos x, f''(0) = 0$$

$$f'''(x) = e^x - e^{-x} + 2 \sin x, f'''(0) = 0$$

$$f^{(4)}(x) = e^x + e^{-x} + 2 \cos x, f^{(4)}(0) = 4 > 0, (n + 1) = 4$$

so $x = 0$ is a local minimum point.

The following theorem is stated under weak conditions then the above one.

Theorem 5.18. Suppose that the function f is continuous on an interval $]x_0 - \delta, x_0 + \delta[$ ($\delta > 0$) and differentiable on $]x_0 - \delta, x_0[$ and on $]x_0, x_0 + \delta[$. Assume that

$$f''(x) \geq 0, (\text{resp.}, \leq 0) \quad \text{on }]x_0 - \delta, x_0[\quad (3)$$

$$f''(x) \leq 0, (\text{resp.}, \geq 0) \quad \text{on }]x_0, x_0 + \delta[\quad (4)$$

Then the function f admits a local maximum (resp. local minimum) at x_0 . Remark that the existence of $f'(x_0)$ isn't mandatory.

Example 5.17. Find the extrema points of the function $f(x) = x^{\frac{1}{3}}(1 - x)^{\frac{2}{3}}$

$$\text{First we calculate derivative: } f'(x) = \frac{\frac{1}{3} - x}{\sqrt[3]{x^2(1-x)}} = 0$$

we can easily check that $x_1 = \frac{1}{3}$ is a stationary point. The derivative at points $x_2 = 0$ et $x_3 = 1$ does not exist. Let $0 < \delta < \frac{1}{3}$, then:

$$f' \left(\frac{1}{3} - \delta \right) > 0, f' \left(\frac{1}{3} + \delta \right) < 0$$

$$f'(-\delta) > 0, f'(\delta) > 0$$

$$f'(1 - \delta) < 0, f'(1 + \delta) > 0$$

So for $x_1 = \frac{1}{3}$ the function admits a maximum, for $x_2 = 0$, there is no extrema. For $x_3 = 1$ the function admits a minimum.

Bounds of a function

Suppose we have to find the maximum (resp, minimum) of a continuous function on an interval $[a, b]$ three cases only are possible:

$$1) x_0 = a \quad 2) x_0 = b \quad 3) x_0 \in]a, b[$$

If $x_0 \in]a, b[$, then the function f admits a local extrema at x_0 , that is a critical point (either stationary or a point such that the derivative does not exist).

If $\{x_1, \dots, x_n\}$, is a finite set, then

$$\begin{aligned} \max_{x \in [a, b]} f(x) &= \max \{f(a), f(b), f(x_1), \dots, f(x_n)\} \\ \min_{x \in [a, b]} f(x) &= \min \{f(a), f(b), f(x_1), \dots, f(x_n)\} \end{aligned}$$

Example 5.18. Find local extrema of the function $f(x) = x^3 - 3x + 3$ on $\left[-3, \frac{3}{2}\right]$

$$\text{We have } f'(x) = 3x^2 - 3 = 0 \Leftrightarrow x_1 = -1, x_2 = 1$$

$$\text{Since } f(-1) = 5, f(1) = 1, f(-3) = -15, f\left(\frac{3}{2}\right) = \frac{15}{8}$$

$$\text{then } \max_{x \in [-3, \frac{3}{2}]} f(x) = 5 \text{ et } \min_{x \in [-3, \frac{3}{2}]} f(x) = -15.$$

Convexity of a curve, point of inflection

There are different types of functions. They are classified according to the categories. One such category is the nature of the graph. Depending upon the nature of the graph, the functions can be divided into two types namely, convex Function and concave Function. Both the concavity and convexity can occur in a function once or more than once. The point where the function is neither concave nor convex is known as inflection point or the point of inflection.

Definition 5.9. If a curve opens in an upward direction or it bends up to make a shape like a cup, it is said to be concave up or convex down. If a curve bends down or resembles a cap, it is known as concave down or convex up. In other words, the tangent lies underneath the curve if the slope of the tangent increases by the increase in an independent variable.

Remark 5.7. If f is a differentiable function, then when $f'' \geq 0$, we have a portion of the graph where the gradient is increasing, so the graph is convex at this section. When $f'' \leq 0$, we have a portion of the graph where the gradient is decreasing, so the graph is concave at this section. .

Definition 5.10. The point of inflection or inflection point is a point in which the concavity of the function changes. It means that the function changes from concave down to concave up or vice versa. In other words, the point in which the rate of change of slope from increasing to decreasing manner or vice versa is known as an inflection point. Those points are certainly not local maxima or minima, but they are stationary points too.

Example 5.19. Let the curve defined by $y = 1 + \sqrt[3]{x}$.

Check the concavity (convexity) at points $A(-1, 0)$, $B(1, 2)$. On a $y''(x) = -\frac{2}{9}x^{-\frac{5}{3}}$, $y''(-1) = \frac{2}{9} > 0$, $y''(1) < 0$, so at point A , the curve is concave up or convex down, and at the point B , it is concave down or convex up.

Corollary 5.3. If x_0 is a point of inflection of a curve $y = f(x)$ and if the second derivative f'' exists at x_0 , then one has necessary $f''(x_0) = 0$.

Theorem 5.19. If f is such that the derivative f''' is continuous at x_0 and $f''(x_0) = 0$ with $f'''(x_0) \neq 0$, then the curve $y = f(x)$ admits a point of inflection at x_0 .

Example 5.20. Consider $y = x^3 - 3x^2 - 9x + 9$. We have $y' = 3x^2 - 6x - 9$, $y'' = 6x - 6$. Then $y''(1) = 0$ and $y'''(1) = 6 \neq 0$. So $x = 1$ is an inflection point.

Remark 5.8. A curve can admit an inflection point at a given point x_0 despite that $f''(x_0)$ does not exist.

Theorem 5.20. Let f be a function such that: $f''(x_0) = \dots = f^{(n)}(x_0) = 0$, $f^{(n+1)}$ is continuous at x_0 and $f^{(n+1)}(x_0) \neq 0$, if n is odd, then the curve $y = f(x)$ is convex up or convex down according to $f^{(n+1)}(x_0) < 0$ or $f^{(n+1)}(x_0) > 0$. If n is even, then x_0 is a point of inflection.

Example 5.21. Let $y = x^5$. we have $y'(x) = 5x^4$,

$$\begin{aligned} y'' &= 20x^3, & y''(0) &= 0 \\ y'''(x) &= 60x^2, & y'''(0) &= 0 \\ y^{(4)}(x) &= 60.2x, & y^{(4)}(0) &= 0 \\ y^{(5)}(x) &= 120, & y^{(5)}(0) &= 120 \neq 0 \\ n &= 4 \text{ even, then } x_0 = 0 \text{ is an inflection point.} \end{aligned}$$

Asymptote of a curve An asymptote is a straight line that constantly approaches a given curve but does not meet at any infinite distance. In other words, Asymptote is a line that a curve approaches as it moves towards infinity. The curves visit these asymptotes but never overtake them.

Definition 5.11. The line $x = a$ is said to be a vertical asymptote of a continuous curve $y = f(x)$ if at least one the two limits $\lim_{x \rightarrow a^+} f(x)$, $\lim_{x \rightarrow a^-} f(x)$ is infinite.

If a curve $y = f(x)$ is defined for $x > A$ (resp. $x < A$), then $y = ax + b$ is said to be oblique asymptote of the curve $y = f(x)$ for $x \rightarrow +\infty$ (resp. $x \rightarrow -\infty$) if $f(x) = ax + b + \alpha(x)$ where $\lim_{x \rightarrow +\infty} \alpha(x) = 0$ (resp. $x \rightarrow -\infty$) (in other words $|f(x) - ax - b|$ is infinitely small with respect to $x \rightarrow +\infty$ (resp. $x \rightarrow -\infty$)).

A horizontal asymptote is a horizontal line, $y = a$, that has the property that either:
 $\lim_{x \rightarrow +\infty} f(x) = a$ or $\lim_{x \rightarrow -\infty} f(x) = a$. This means, that as x approaches positive or negative infinity, the function tends to a constant value a .

Example 5.22. Consider $y = \frac{8}{x-2}$. The line $x = 2$ is a vertical asymptote because
 $\lim_{x \rightarrow 2^+} \frac{8}{x-2} = +\infty$, $\lim_{x \rightarrow 2^-} \frac{8}{x-2} = -\infty$.

Example 5.23. Let $y = \frac{x^2 + x}{x-1}$. Since $f(x) = x + 2 + \frac{2}{x-1}$ et $\lim_{x \rightarrow \infty} \frac{2}{x-1} = 0$, then the line $y = x + 2$ is an oblique asymptote of the curve $y = \frac{x^2 + x}{x-1}$ for $x \rightarrow +\infty$ and for $(x \rightarrow -\infty)$.

Theorem 5.21. Let $y = f(x)$ be a given curve. $y = f(x)$ admits an oblique asymptote of the form $y = ax + b$ for $x \rightarrow +\infty$ ($x \rightarrow -\infty$) if and only if the following limits

$$\lim_{\substack{x \rightarrow +\infty \\ (x \rightarrow -\infty)}} \frac{f(x)}{x} = a, \quad \lim_{\substack{x \rightarrow +\infty \\ (x \rightarrow -\infty)}} [f(x) - ax] = b$$

exist.

Remark 5.9. The existence of the limits

$$\lim_{\substack{x \rightarrow +\infty \\ (x \rightarrow -\infty)}} \frac{f(x)}{x} = a, \quad \lim_{\substack{x \rightarrow +\infty \\ (x \rightarrow -\infty)}} [f(x) - ax] = b$$

is necessary. Indeed for the curve

$y = \sqrt{x}$ ($x \geq 0$) we have $\lim_{x \rightarrow +\infty} \frac{\sqrt{x}}{x} = 0 = a$ and $\lim_{x \rightarrow +\infty} [\sqrt{x} - 0x] = +\infty$, which means $b = +\infty$, so this curve does not have asymptote.

Example 5.24. Let the curve defined by $y = xe^{\frac{1}{x^2}}$

1) Vertical asymptote

$$\lim_{x \rightarrow 0^+} xe^{\frac{1}{x^2}} = \lim_{y \rightarrow +\infty} \frac{e^{y^2}}{y} = +\infty, \quad \lim_{x \rightarrow 0^-} xe^{\frac{1}{x^2}} = \lim_{y \rightarrow -\infty} \frac{e^{y^2}}{y} = -\infty.$$

then $x = 0$ is a vertical asymptote.

2) oblique asymptote

$$a = \lim_{x \rightarrow \pm\infty} x \frac{e^{\frac{1}{x^2}}}{x} = \lim_{x \rightarrow \pm\infty} x e^{\frac{1}{x^2}} = 1,$$

$$b = \lim_{x \rightarrow \pm\infty} \left[x e^{\frac{1}{x^2}} - x \right] = \lim_{x \rightarrow \pm\infty} \left[x \left(1 + \frac{1}{x^2} \right) - x \right] = 0.$$

thus $b = 0$. So $y = x$ is an oblique asymptote of the considered curve.

5.2 Convex Functions

5.2.1 Introduction

Convexity is the study of convex sets and functions, it constitutes a branch of geometry and analysis that unites phenomena that are at first sight totally dissimilar. It intervenes at various levels in very varied branches of mathematics (number theory, combinatorial problems, optimization, functional analysis, etc.). In analysis, the property of convexity is relatively sought after, since as we will see in the following, it offers a good number of nice properties on the function.

Definition 5.12. A function $f : I \longrightarrow \mathbb{R}$ is said to be convex when: $\forall (x_1, x_2) \in I, \forall \lambda \in [0, 1]$

$$f(\lambda x_1 + (1 - \lambda)x_2) \leq \lambda f(x_1) + (1 - \lambda)f(x_2)$$

this means that for all x_1 and x_2 of I , the line segment connecting the points $(x_1, f(x_1))$ and $(x_2, f(x_2))$ are located above the curve representing f .

Definition 5.13. f will be said to be concave if $-f$ is convex i.e. $\forall (x_1, x_2) \in I, \forall \lambda \in [0, 1]$

$$f(\lambda x_1 + (1 - \lambda)x_2) \geq \lambda f(x_1) + (1 - \lambda)f(x_2)$$

Definition 5.14. A function $f : I \longrightarrow \mathbb{R}$ is said to be strictly convex if

$$\forall (x_1, x_2) \in I, x_1 \neq x_2, \forall \lambda \in [0, 1]$$

$$f(\lambda x_1 + (1 - \lambda)x_2) < \lambda f(x_1) + (1 - \lambda)f(x_2)$$

Definition 5.15. A function $f : I \longrightarrow \mathbb{R}$ is said to be strictly concave if

$$\forall (x_1, x_2) \in I, x_1 \neq x_2, \forall \lambda \in [0, 1]$$

$$f(\lambda x_1 + (1 - \lambda)x_2) > \lambda f(x_1) + (1 - \lambda)f(x_2)$$

Examples:

1. The function $f : x \longrightarrow x^2$ is convex on $\mathbb{R}$

Indeed for everyone x, y of I and for $\lambda \in [0, 1]$ we have:

$$\begin{aligned}
& [\lambda x + (1 - \lambda) y]^2 \leq \lambda x^2 + (1 - \lambda) y^2 \\
& \iff \\
& \lambda^2 x^2 + 2\lambda(1 - \lambda)xy + (1 - \lambda)^2 y^2 \leq \lambda x^2 + (1 - \lambda) y^2 \\
& \iff \\
& \lambda^2 [-x^2 + 2xy - y^2] - \lambda [-x^2 + 2xy - y^2] \leq 0 \\
& \iff \\
& \lambda(1 - \lambda) [-x^2 + 2xy - y^2] \leq 0 \\
& \iff \\
& -\lambda(1 - \lambda) (x - y)^2 \leq 0 \text{ which is true}
\end{aligned}$$

2. Affine functions are both convex and concave.
3. La fonction $\exp$ est strictement convexe sur $\mathbb{R}$.
4. La fonction $\log$ est strictement concave sur $\mathbb{R}^+$.

5.2.2 Properties of convex functions

Proposition 5.1. *Let $f : I \rightarrow \mathbb{R}$ be a function on the interval I . The following assertions are equivalent.*

1. f is convex on I ,
2. For any collection of positive real numbers $\alpha_1, \alpha_2, \dots, \alpha_n$ with $\alpha_1 + \alpha_2 + \dots + \alpha_n = 1$ and any collection of real numbers $x_1, x_2, \dots, x_n$ in I we have

$$f(\alpha_1 x_1 + \alpha_2 x_2 + \dots + \alpha_n x_n) \leq \alpha_1 f(x_1) + \alpha_2 f(x_2) + \dots + \alpha_n f(x_n).$$

Proof. The implication **2** $\Rightarrow$ **1** is clear. The implication **1** $\Rightarrow$ **2** is obtained by induction. If $n = 2$, then $\alpha_2 = 1 - \alpha_1$ and the convexity of f leads to

$$f(\alpha_1 x_1 + \alpha_2 x_2) \leq \alpha_1 f(x_1) + \alpha_2 f(x_2)$$

Suppose that it is true at the rank n and let us prove that it is true at the rank $n + 1$.

$$\begin{aligned}
& f(\alpha_1 x_1 + \alpha_2 x_2 + \cdots + \alpha_n x_n + \alpha_{n+1} x_{n+1}) \\
&= f\left(\left(\sum_{j=1}^{j=n} \alpha_j\right) \left(\sum_{k=1}^{k=n} \frac{\alpha_k}{\left(\sum_{j=1}^{j=n} \alpha_j\right)} x_k\right) + \alpha_{n+1} x_{n+1}\right) \\
&\leq \left(\sum_{j=1}^{j=n} \alpha_j\right) f\left(\sum_{k=1}^{k=n} \frac{\alpha_k}{\left(\sum_{j=1}^{j=n} \alpha_j\right)} x_k\right) + \alpha_{n+1} f(x_{n+1}) \\
&\leq \left(\sum_{j=1}^{j=n} \alpha_j\right) \sum_{k=1}^{k=n} \frac{\alpha_k}{\left(\sum_{j=1}^{j=n} \alpha_j\right)} f(x_k) + \alpha_{n+1} f(x_{n+1}) \\
&= \alpha_1 f(x_1) + \alpha_2 f(x_2) + \cdots + \alpha_n f(x_n) + \alpha_{n+1} f(x_{n+1})
\end{aligned}$$

□

Theorem 5.22. *Let $f : I \rightarrow \mathbb{R}$ be a function defined on the interval I . The following propositions are equivalents*

1. f is convex on I ,
2. For all $x, y, z \in I, x < y < z$ implies $\frac{f(y)-f(x)}{y-x} \leq \frac{f(z)-f(x)}{z-x}$,
3. For all $x, y, z \in I, x < y < z$ implies $\frac{f(y)-f(x)}{y-x} \leq \frac{f(z)-f(y)}{z-y}$,
4. For all $x, y, z \in I, x < y < z$ implies $\frac{f(z)-f(x)}{z-x} \leq \frac{f(z)-f(y)}{z-y}$.

Proof. We prove **1** $\Leftrightarrow$ **2**, the other equivalences are checked similarly.

- **1** $\Rightarrow$ **2**. For $x < y < z$ we write $y = \mu x + (1 - \mu)z$ where $\mu = \frac{z-y}{z-x} \in [0, 1]$. From the convexity of f we deduce

$$f(y) \leq \mu f(x) + (1 - \mu)f(z) = \frac{z-y}{z-x} f(x) + \frac{y-x}{z-x} f(z).$$

Adding $-f(x)$ and dividing by $(y - x)$ we get

$$\frac{f(y) - f(x)}{y - x} \leq \frac{f(z) - f(x)}{z - x}$$

- **2** $\Rightarrow$ **1**. Let $x_1, x_2 \in I$ with $x_1 < x_2$ and let $\lambda \in [0, 1]$. It is clear that if $\lambda = 0$ or $\lambda = 1$ then $f(\lambda x_1 + (1 - \lambda)x_2) \leq \lambda f(x_1) + (1 - \lambda)f(x_2)$.

If $\lambda \in (0, 1)$ then setting $x = x_1, y = \lambda x_1 + (1 - \lambda)x_2$ and $z = x_2$ we get $x < y < z$ and consequently

$$\frac{f(y) - f(x)}{y - x} \leq \frac{f(z) - f(x)}{z - x}$$

Leading to

$$f(y) \leq \frac{z-y}{z-x}f(x) + \frac{y-x}{z-x}f(z),$$

that is

$$f(\lambda x_1 + (1-\lambda)x_2) \leq \lambda f(x_1) + (1-\lambda)f(x_2)$$

□

Definition 5.16. Given $f : I \longrightarrow \mathbb{R}$ and $\alpha \in I$, the application $\varphi_\alpha : I \setminus \{\alpha\} \longrightarrow \mathbb{R}$ defined by $\varphi_\alpha = \frac{f(x)-f(\alpha)}{x-\alpha}$, is called the rate of increase.

The following corollary is an equivalent version of the above Theorem.

Proposition 5.2. A function $f : I \longrightarrow \mathbb{R}$ is convex if and only if for all $\alpha \in I$ the function φ_α is increasing.

Proof. Suppose that f is convex, let us show the growth of φ_α .

Let $(x, y, z) \in I^3$ be such that $x < y$ and $z \in [x, y]$, there exists $\lambda \in [0, 1]$ such that $z = \lambda x + (1-\lambda)y$

Indeed we can éwrite $\lambda = \frac{y-z}{y-x}$, so

$$\begin{aligned} f(z) &\leq \lambda f(x) + (1-\lambda)f(y) && \iff f(z) - f(y) \leq \lambda(f(x) - f(y)) \\ \iff f(z) - f(y) &\leq \frac{y-z}{y-x}(f(x) - f(y)) && \iff \frac{f(y) - f(z)}{y-z} \geq \frac{f(y) - f(x)}{y-x} \\ \iff \varphi_x(y) &\leq \varphi_z(y) \text{ because } y - z < 0, \end{aligned}$$

on the other hand $1 - \lambda = \frac{z-x}{y-x}$ then

$$\begin{aligned} f(z) &\leq \lambda f(x) + (1-\lambda)f(y) && \iff f(z) - f(x) \leq (1-\lambda)(f(y) - f(x)) \\ \iff f(z) - f(x) &\leq \frac{z-x}{y-x}(f(y) - f(x)) && \iff \frac{f(z) - f(x)}{z-x} \leq \frac{f(y) - f(x)}{y-x} \\ \iff \varphi_x(z) &\leq \varphi_x(y) \text{ because } z - x > 0, \end{aligned}$$

hence the growth of φ_α .

During the proof, we verified the inequality of the three slopes. □

Conversely. Suppose that the application φ_α is increasing for all $\alpha \in I$, let us show the convexity of f .

Let $x, y, z \in I^3$ with $x < z < y$ and $\lambda \in [0, 1]$.

We set $z = \lambda x + (1-\lambda)y$, such that $\lambda = \frac{y-z}{y-x}$ and $1-\lambda = \frac{z-x}{y-x}$

By growth of φ_α we have on the one hand

$$\begin{aligned}
\varphi_x(y) \leq \varphi_z(y) &\iff \frac{f(y) - f(x)}{y - x} \leq \frac{f(y) - f(z)}{y - z} \iff f(y) - f(z) \geq \frac{y - z}{y - x} (f(y) - f(x)) \\
&\iff f(z) - f(y) \leq \frac{y - z}{y - x} (f(x) - f(y)) \text{ because } y - z < 0 \iff f(z) - f(y) \leq \lambda (f(x) - f(y)) \\
&\iff f(z) \leq \lambda f(x) + (1 - \lambda) f(y) \iff f(\lambda x + (1 - \lambda) y) \leq \lambda f(x) + (1 - \lambda) f(y)
\end{aligned}$$

On the other hand

$$\begin{aligned}
\varphi_x(z) \leq \varphi_x(y) &\iff \frac{f(z) - f(x)}{z - x} \leq \frac{f(y) - f(x)}{y - x} \\
&\iff f(z) - f(x) \leq \frac{z - x}{y - x} (f(y) - f(x)) \text{ because } z - x > 0 \\
&\iff f(z) - f(x) \leq (1 - \lambda) (f(y) - f(x)) \\
&\iff f(z) \leq \lambda f(x) + (1 - \lambda) f(y) \\
&\iff f(\lambda x + (1 - \lambda) y) \leq \lambda f(x) + (1 - \lambda) f(y),
\end{aligned}$$

Hence the convexity of f . □

Theorem 5.23 (Jensen's inequality). *Let f be a convex function defined from I into $\mathbb{R}$. For any $(x_1, x_2, \dots, x_n) \in I$ and for any family $(\lambda_1, \lambda_2, \dots, \lambda_n) \in \mathbb{R}$ such that $\sum_{i=1}^n \lambda_i = 1$ we have the following convexity inequality:*

$$f\left(\sum_{i=1}^n \lambda_i x_i\right) \leq \sum_{i=1}^n \lambda_i f(x_i)$$

Proof. The proof is done by recurrence for all $n \in \mathbb{N}$

For $n = 2$, we have by convexity

$$f(\lambda_1 x_1 + \lambda_2 x_2) \leq \lambda_1 f(x_1) + \lambda_2 f(x_2)$$

Let us suppose that the inequality is true for $n \in \mathbb{N} \setminus \{1, 2\}$, let us show that this implies that it is true for $n + 1$.

Let $(x_1, x_2, \dots, x_{n+1}) \in I$ and $(\lambda_1, \lambda_2, \dots, \lambda_{n+1}) \in \mathbb{R}$ be such that $\sum_{i=1}^{n+1} \lambda_i = 1$

If $\sum_{i=1}^n \lambda_i = 0$, then $\lambda_1 = \lambda_2 = \dots = \lambda_n = 0$ and $\lambda_{n+1} = 1$, inequality is true.

Now let $\sum_{i=1}^n \lambda_i = S$, we can write

$$\sum_{i=1}^{n+1} \lambda_i x_i = S \frac{\sum_{i=1}^n \lambda_i x_i}{S} + \lambda_{n+1} x_{n+1}, \text{ with } S \text{ and } \lambda_{n+1} \text{ are positive, with sum equal to 1.}$$

By the convexity of f

$$f\left(\sum_{i=1}^{n+1} \lambda_i x_i\right) \leq S \times f\left(\frac{\sum_{i=1}^n \lambda_i x_i}{S}\right) + \lambda_{n+1} f(x_{n+1})$$

Now, if we put for each $1 \leq i \leq n$, $\lambda'_i = \frac{\lambda_i}{S}$, $\sum_{i=1}^n \lambda'_i = 1$

So, by hypothesis of recurrence we have

$$f\left(\sum_{i=1}^n \frac{\lambda_i}{S} x_i\right) \leq \sum_{i=1}^n \frac{\lambda_i}{S} f(x_i).$$

Eventually

$$f\left(\sum_{i=1}^{n+1} \lambda_i x_i\right) \leq \underbrace{S \times \sum_{i=1}^n \frac{\lambda_i}{S} f(x_i) + \lambda_{n+1} f(x_{n+1})}_{\sum_{i=1}^{n+1} \lambda_i f(x_i)}$$

which ends the proof. □

5.2.3 Continuous convex functions

Definition 5.17. A function $f : I \longrightarrow \mathbb{R}$ is continuous at a point α of I if:

$$\forall \varepsilon > 0, \exists \eta > 0, \forall x \in I; |x - \alpha| < \eta \implies |f(x) - f(\alpha)| < \varepsilon$$

Theorem 5.24. If the function $f : I \longrightarrow \mathbb{R}$ is convex on I , then it admits a right derivative $\dot{a}$ and a left derivative $\dot{a}$ at any point in the interior of I . *noté $\dot{I}$.* f is continuous on $\dot{I}$.

Proof. Let the application be $\varphi_\alpha : I \setminus \{\alpha\} \longrightarrow \mathbb{R}, \forall \alpha \in I$ for all $x \in]\alpha, \alpha[$ and $y \in]\alpha, b[$

Then the limit à left and à right in α of φ_α exists:

$$\begin{aligned}\lim_{x \rightarrow \alpha^-} \varphi_\alpha(x) &= \lim_{x \rightarrow \alpha^-} \frac{f(x) - f(\alpha)}{x - \alpha} = f'_g(\alpha) \\ \lim_{y \rightarrow \alpha^+} \varphi_\alpha(y) &= \lim_{y \rightarrow \alpha^+} \frac{f(y) - f(\alpha)}{y - \alpha} = f'_d(\alpha)\end{aligned}$$

hence the continuity of f .

Let $\alpha \in \overset{\circ}{I}$ then $\forall x \in]a, \alpha[$ and $y \in]\alpha, b[$ on has

$$\frac{f(x) - f(\alpha)}{x - \alpha} \leq \frac{f(y) - f(\alpha)}{y - \alpha}$$

When x tends to α^- and y tends to α^+ , we have inequality

$$f'_g(\alpha) \leq f'_d(\alpha)$$

Let $\beta \in I^0$, with $\alpha < \beta, \forall x \in]\alpha, \beta[$ we have:

$$\begin{aligned}\frac{f(x) - f(\alpha)}{x - \alpha} &\leq \frac{f(\beta) - f(\alpha)}{\beta - \alpha} \leq \frac{f(x) - f(\beta)}{x - \beta} \\ \Rightarrow \lim_{x \rightarrow \alpha^-} \frac{f(x) - f(\alpha)}{x - \alpha} &\leq \frac{f(\beta) - f(\alpha)}{\beta - \alpha} \leq \lim_{x \rightarrow \alpha^-} \frac{f(x) - f(\beta)}{x - \beta} \\ \Leftrightarrow f'_g(\alpha) &\leq \frac{f(\beta) - f(\alpha)}{\beta - \alpha} \leq f'_g(\beta)\end{aligned}$$

SO

$$f'_g(\alpha) \leq f'_g(\beta)$$

hence the growth of f'_g on $\overset{\circ}{I}$

Let $\beta \in I^0$, with $\alpha < \beta, \forall y \in]\alpha, \beta[$ we have:

$$\begin{aligned}\frac{f(y) - f(\alpha)}{y - \alpha} &\leq \frac{f(\beta) - f(\alpha)}{\beta - \alpha} \leq \frac{f(y) - f(\beta)}{y - \beta} \\ \Rightarrow \lim_{y \rightarrow \alpha^+} \frac{f(y) - f(\alpha)}{y - \alpha} &\leq \frac{f(\beta) - f(\alpha)}{\beta - \alpha} \leq \lim_{y \rightarrow \alpha^+} \frac{f(y) - f(\beta)}{y - \beta} \\ \Leftrightarrow f'_d(\alpha) &\leq \frac{f(\beta) - f(\alpha)}{\beta - \alpha} \leq f'_d(\beta).\end{aligned}$$

So

$$f'_d(\alpha) \leq f'_d(\beta)$$

hence the growth of f'_d on $\overset{\circ}{I}$.

□

Remark 5.10. *If f is continuous on $I \nRightarrow f$ is convex on I*

If f is derivable on $I \nRightarrow f$ is convex on I

5.2.4 Convex derivable functions

Definition 5.18. *A function $f : I \rightarrow \mathbb{R}$ is derivable in α of $\overset{\circ}{I}$ if and only if:*

$$\lim_{x \rightarrow \alpha} \frac{f(x) - f(\alpha)}{x - \alpha} \text{ exists}$$

Definition 5.19. *Let f be a function defined on the open interval I in $\mathbb{R}$. Let α be a point of I . We say that α is a:*

- local maximum of f if

$$\exists \varepsilon > 0, |x - \alpha| \leq \varepsilon \implies f(x) \leq f(\alpha)$$

- local minimum of f if

$$\exists \varepsilon > 0, |x - \alpha| \leq \varepsilon \implies f(x) \geq f(\alpha)$$

Proposition 5.3. *Let f be a function defined on the open interval I in $\mathbb{R}$. If f has a local extremum (maximum or minimum) at a point α of I and if f is differentiable at α , then $f'(\alpha) = 0$.*

Proof. If α is a local minimum of f , then it is a local maximum of $-f$, without loss of generality we can assume that α is a local maximum.

$$\exists \varepsilon > 0, \forall x \in [\alpha - \varepsilon, \alpha + \varepsilon] : f(x) \leq f(\alpha)$$

so for all x in $[\alpha - \varepsilon, \alpha[$

$$\frac{f(x) - f(\alpha)}{x - \alpha} \geq 0 \text{ thus } \lim_{x \rightarrow \alpha^-} \frac{f(x) - f(\alpha)}{x - \alpha} = f'(\alpha) \geq 0$$

for all x in $] \alpha, \alpha + \varepsilon]$

$$\frac{f(x) - f(\alpha)}{x - \alpha} \leq 0 \text{ so } \lim_{x \rightarrow \alpha^+} \frac{f(x) - f(\alpha)}{x - \alpha} = f'(\alpha) \leq 0$$

which implies that $f'(\alpha) = 0$. □

Theorem 5.25. *Let a function $f : I \rightarrow \mathbb{R}$ be differentiable and convex on I then $\forall \alpha \in I, \forall x \in I$*

$$f(x) \geq f(\alpha) + f'(\alpha)(x - \alpha)$$

Proof. Let f be a convex and differentiable function on I , or $\alpha \in I$, for $x \in I$, let

$$g(x) = f(x) - f(\alpha) - f'(\alpha)(x - \alpha)$$

then g is differentiable on I and:

$$\forall x \in I, g'(x) = f'(x) - f'(\alpha).$$

or, f' is increasing on I , we therefore obtain the table of following variations:

x	$x \leq \alpha$	$x \geq \alpha$
$g'(x)$	−	+
$g(x)$	↘	↗

which proves that $\forall x \in I, g(x) \geq 0$, so $\forall x \in I, f(x) \geq f(\alpha) + f'(\alpha)(x - \alpha)$ □

Proposition 5.4. *Let $f : I \rightarrow \mathbb{R}$ a continuous and derivable function on I , then it is convex if and only if f' is increasing.*

Proof. Let us start with the necessary condition (growth of f'). Let $x, y \in I$ be two points such that $x < y$. For all $z \in [x, y]$

$$\frac{f(z) - f(x)}{z - x} \leq \frac{f(z) - f(y)}{z - y}$$

making z tend towards x

$$f'(x) \leq \frac{f(x) - f(y)}{x - y}$$

similarly, making z tend towards y

$$\frac{f(x) - f(y)}{x - y} \leq f'(y)$$

so $f'(x) \leq f'(y)$.

Let us now show the sufficient condition (convexity of f).

For $\lambda \in [0, 1]$ and $a = \lambda x + (1 - \lambda)y$. Let us apply Mean value theorem on the two intervals $[x, a]$ and $[a, y]$, there exist $c_1 \in]x, a[$ and $c_2 \in]a, y[$ such that :

$$\frac{f(x) - f(a)}{x - a} = f'(c_1) \text{ and } \frac{f(y) - f(a)}{y - a} = f'(c_2)$$

the function f' being increasing, we therefore have

$$\frac{f(x) - f(a)}{x - a} \leq \frac{f(y) - f(a)}{y - a},$$

this leads to $f(a) \leq \lambda f(x) + (1 - \lambda)f(y)$. □

Corollary 5.4. *Let the function $f : I \rightarrow \mathbb{R}$ be twice differentiable and continuous on an interval I , then it is convex if $f''(x) > 0$ for all x in I .*

Proof. This is an immediate consequence of the proposition proved (1.3).

$$f \text{ convex on } I \iff f' \text{ is increasing on } I \iff f'' \geq 0 \text{ on } I$$

□

Example 5.25. *Case of power functions $f : x \mapsto x^h$ for $h \in]0, +\infty[$. f is of class C^∞ and we have:*

$$f'(x) = hx^{h-1} \text{ increasing } \forall x > 0$$

And

$$f''(x) = h(h-1)x^{h-2} \geq 0 \quad \forall x > 0$$

hence

$$f \text{ convex} \iff h \leq 0 \text{ or } h \geq 1$$

$$f \text{ concave} \iff h \in [0, 1]$$